

210mm

297mm

pharmaniaga[®]
Noradrenaline
1mg/mL Injection



COMPOSITION

Each mL contains 2mg of Noradrenaline Bitartrate equivalent to 1mg Noradrenaline.

PRODUCT DESCRIPTION

Clear, odourless liquid and free from any precipitate. The compatible infusion solutions for intravenous use are as follows:

- 5% Dextrose Injection.
 - 0.9% Sodium Chloride and 5% Dextrose Injection.
- If not required immediately, the diluted solution may be stored up to 24 hours at temperature below 30°C after mixing with the compatible solutions.

PHARMACODYNAMICS

Pharmacotherapeutic group: Adrenergic and dopaminergic agents. ATC code: C01CA03. The vascular effects in the doses normally used clinical result from the simultaneous stimulation of alpha- and beta-adrenergic receptors in the heart and vascular system. Except in the heart, its action is predominantly on the alpha receptors. This results in an increase in the force (and in the absence of vagal inhibition, in the rate) of myocardial contraction. Peripheral resistance increases and diastolic and systolic pressures are raised.

The increase in blood pressure may cause a reflex decrease in heart rate. Vasoconstriction may result in decreased blood flow in kidneys, liver, skin and smooth muscles. Local vasoconstriction may cause haemostasis and/or necrosis. The effect on blood pressure disappears 1-2 minutes after stopping the infusion.

PHARMACOKINETICS

Two stereoisomers of Noradrenaline exist, the biologically active L-isomer is the one present in Noradrenaline (Norepinephrine) Concentrate.

Absorption

- Subcutaneous: Poor
- Oral: Noradrenaline is rapidly inactivated in the gastro-intestinal tract following oral administration.
- After intravenous administration: Noradrenaline has a plasmatic half-life of about 1 to 2 minutes.

Distribution

- Noradrenaline is rapidly cleared from plasma by a combination of cellular reuptake and metabolism. It does not readily cross the blood-brain barrier.

Biotransformation

- Methylation by catechol-o-methyltransferase.
- Deamination by monoamine oxidase (MAO).
- Ultimate metabolites from both is 4- hydroxy-3- methoxymandelic acid.
- Intermediate metabolites include normetanephrine and 3,4- dihydroxymandelic acid.

Elimination

- Noradrenaline is mainly eliminated as glucuronide or sulphate conjugates of the metabolites in the urine.

INDICATION

For blood pressure control in certain acute hypotensive states (e.g. pheochromocytectomy, sympathectomy, poliomyelitis, spinal anesthesia, myocardial infarction, septicaemia, blood transfusion and drug interactions). As an adjunct in the treatment of cardiac arrest and profound hypotension.

RECOMMENDED DOSAGE

Noradrenaline Injection contains noradrenaline as the acid tartrate. It is a concentrated, potent drug which must be diluted in dextrose containing solutions prior to infusion. An infusion of noradrenaline should be given into a large vein.

Dilution instructions

Either add 2 ml concentrate to 48 ml glucose 5% solution (or sodium chloride 9 mg/ml (0.9%) with glucose 5% solution) for administration by syringe pump, or add 20 ml of concentrate to 480 ml glucose 5 % solution (or sodium chloride 9 mg/ml (0.9%) with glucose 5% solution) for administration by drip counter. In both cases the final concentration of the infusion solution is 40 mg/litre noradrenaline base (which is equivalent to 80 mg/litre noradrenaline tartrate). Dilutions other than 40 mg/litre noradrenaline base may also be used (refer Initial rate of infusion). If dilutions other than 40 mg/litre noradrenaline base are used, check the infusion rate calculation carefully before starting treatment. The product is compatible with PVC infusion bags. Any unused medicinal product or waste material should be disposed of in accordance with local requirements.

Adults

Initial rate of infusion:

When diluted as recommended above (the concentration of the prepared infusion is 40 mg/litre noradrenaline base (80 mg/litre noradrenaline tartrate)) the initial rate of infusion, at a body weight of 70 kg, should be between 10 ml/hour and 20 ml/hour (0.16 to 0.33 ml/min). This is equivalent to 0.4 mg/hour to 0.8 mg/hour noradrenaline base (0.8 mg/hour to 1.6 mg/hour noradrenaline tartrate). At a lower initial infusion rate of 5 ml/hour (0.08 ml/min), equivalent to 0.2 mg/hour noradrenaline base (0.4 mg/hour noradrenaline tartrate).

Titration of dose:

Once an infusion of noradrenaline has been established the dose should be titrated in steps of 0.05 -0.1µg/kg/min of noradrenaline base according to the pressor effect observed. There is great individual variation in the dose required to attain and maintain normotension. The aim

should be to establish a low normal systolic blood pressure (100 - 120 mm Hg) or to achieve an adequate mean arterial blood pressure (greater than 65 - 80 mm Hg – depending on the patient's condition).

Noradrenaline Infusion Solution 40mg mg/litre (40µg/ml) noradrenaline base			
Patient's Weight	Posology (µg/kg/min) noradrenaline base	Posology (mg/min) noradrenaline base	Infusion Rate (ml/hour)
50 kg	0.05	0.15	3.75
	0.1	0.3	7.5
	0.25	0.75	18.75
	0.5	1.5	37.5
	1	3	75
60 kg	0.05	0.18	4.5
	0.1	0.36	9
	0.25	0.9	22.5
	0.5	1.8	45
	1	3.6	90
70 kg	0.05	0.21	5.25
	0.1	0.42	10.5
	0.25	1.05	26.25
	0.5	2.1	52.5
	1	4.2	105
80 kg	0.05	0.24	6
	0.1	0.48	12
	0.25	1.2	30
	0.5	2.4	60
	1	4.8	120
90 kg	0.05	0.27	6.75
	0.1	0.54	13.5
	0.25	1.35	33.75
	0.5	2.7	67.5
	1	5.4	135

Some clinicians may prefer to dilute to other concentrations. If dilutions other than 40 mg/L are used, check the infusion rate calculation carefully before starting treatment.

Renal or hepatic impairment:

There is no experience in treatment of renally or hepatically impaired patients.

Elderly:

As for adults **but** refer Warning and Precautions.

Restoration of Blood Pressure in Acute Hypotensive States:

Blood volume depletion should always be corrected as fully as possible before any vasopressor is administered. When, as an emergency measure, intra-aortic pressures must be maintained to prevent cerebral or coronary artery ischaemia, noradrenaline can be administered before and concurrently with blood volume replacement.

Diluent: Noradrenaline Injection should be diluted in 5% dextrose in water or 5% dextrose in normal saline solution. These dextrose containing fluids provide protection against significant loss of potency due to oxidation. Administration in saline solution alone is not recommended. Whole blood or plasma, if indicated to

increase blood volume, should be administered separately (for example, by use of a Y-tube and individual containers if given simultaneously).

Average Dosage: Add 4mL of Noradrenaline Injection (4mg of Noradrenaline) to 1000ml of a 5% dextrose containing solution. Each mL of this dilution contains 4mcg of the base of noradrenaline (or 8mcg of the acid tartrate). Give this solution by intravenous infusion. Insert a plastic intravenous catheter through a suitable bore needle well advanced centrally into the vein and securely fixed with adhesive tape, avoiding, if possible, a catheter tie-in technique as this promotes stasis. An I.V. drip chamber or other suitable metering device is essential to permit an accurate estimation of the rate of flow in drops per minute. After observing the response to an initial dose of 2mL to 3mL (from 8mcg to 12mcg of base) per minute, adjust the rate of flow to establish and maintain a low normal blood pressure (usually 80 mmHg to 100 mmHg systolic) sufficient to maintain the circulation to vital organs. In previously hypertensive patients, it is recommended that the blood pressure should be raised no higher than 40mmHg below the pre-existing systolic pressure. The average maintenance dose ranges from 0.5mL to 1mL per minute (from 2mcg to 4mcg of base).

High Dosage: Great individual variation occurs in the dose required to attain and maintain an adequate blood pressure. In all cases, dosage of noradrenaline should be titrated according to the response of the patient. Occasionally much larger or even enormous daily doses (as high as 68mg base or 17 ampoules) may be necessary if the patient remains hypotensive, but occult blood volume depletion should always be suspected and corrected when present. Central venous pressure monitoring is usually helpful in detecting and treating this situation.

Fluid Intake: The degree of dilution depends on clinical fluid volume requirements. If large volumes of fluid (dextrose) are needed at a flow rate that would involve an excessive dose of the pressor agent per unit of time, a solution more dilute than 4mcg per mL should be used. On the other hand, when large volumes of fluid are clinically undesirable, a concentration greater than 4mcg per mL may be necessary.

Duration of Therapy: The infusion should be continued until adequate blood pressure and tissue perfusion are maintained without therapy. Infusions of noradrenaline should be reduced gradually, avoiding abrupt withdrawal. In some of the reported cases of vascular collapse due to acute myocardial infarction, treatment was required for up to six days.

Adjunctive Treatment in Cardiac Arrest:

Infusions of noradrenaline are usually administered intravenously during cardiac resuscitation to restore and maintain an adequate blood pressure after an effective heartbeat and ventilation have been established by other means. [Noradrenaline's powerful beta-adrenergic stimulating action is also thought to increase the strength and effectiveness of systolic contractions once they occur.]

Average Dosage: To maintain systemic blood pressure during the management of cardiac arrest, noradrenaline is used in the same manner as described under Restoration of

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01	23.08.19	- Replace PM code to 2003255.
02	27.08.19	New Word file received.
03	07.10.2020	Amend PM code, co. no. and revision date.

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Blood Pressure in Acute Hypotensive States. Parenteral drug products should be inspected visually for particulate matter and discolouration prior to use, whenever solution and container permit. Do not use the solution if its colour is pinkish or darker than slightly yellow or if it contains a precipitate. Avoid contact with alkalis and oxidising agents.

ROUTE OF ADMINISTRATION
For intravenous (IV) infusion only.

Method of administration:
Administer as a diluted solution via a central venous catheter. The infusion should be at a controlled rate using either a syringe pump or an infusion pump or a drip counter. For instructions on dilution of the medicinal product before administration, refer Recommended Dosage.

CONTRAINDICATIONS
Noradrenaline should not be given to patients who are hypotensive from blood volume deficits except as an emergency measure to maintain coronary and cerebral artery perfusion until blood volume replacement therapy can be completed. If noradrenaline is continuously administered to maintain blood pressure in the absence of blood volume replacement, the following may occur: severe peripheral and visceral vasoconstriction, decreased renal perfusion and urine output, poor systemic blood flow despite “normal” blood pressure, tissue hypoxia, and lactate acidosis. Noradrenaline should also not be given to patients with mesenteric or peripheral vascular thrombosis (because of the risk of increasing ischaemia and extending the area of infarction) unless, in the opinion of the attending physician, the administration of noradrenaline is necessary as a life-saving procedure. Cyclopropane and halothane anaesthetics increase cardiac autonomic irritability and therefore seem to sensitise the myocardium to the action of intravenously administered adrenaline or noradrenaline. Hence, the use of noradrenaline during cyclopropane and halothane anaesthesia is generally considered contraindicated because of the risk of producing ventricular tachycardia or fibrillation. The same type of cardiac arrhythmias may result from the use of noradrenaline in patients with profound hypoxia or hypercarbia.

WARNINGS AND PRECAUTIONS
Warnings
Noradrenaline Injection should be used with extreme caution in patients receiving monoamine oxidase inhibitors (MAOI) or antidepressants of the triptyline or imipramine types, because severe, prolonged hypertension may result.

Noradrenaline should only be administered by healthcare professionals who are familiar with its use. Elderly patients may be especially sensitive to the effects of noradrenaline. Particular caution should be observed in patients with hypotension following myocardial infarction, in patients with Prinzmetal's variant angina and in patients with diabetes, hypertension or hyperthyroidism.

Noradrenaline should be used with caution in patients who exhibit profound hypoxia or hypercarbia. Prolonged administration of any potent vasopressor may result in plasma volume depletion which should be continuously

corrected by appropriate fluid and electrolyte replacement therapy. If plasma volumes are not corrected, hypotension may recur when the infusion is discontinued, or blood pressure may be maintained at the risk of severe peripheral and visceral vasoconstriction (e.g., decreased renal perfusion) with diminution in blood flow and tissue perfusion with subsequent tissue hypoxia and lactic acidosis and possible ischaemic injury. Noradrenaline Injection contains sodium, thus to be taken into consideration by patients on a controlled sodium diet.

Precautions
General
Avoid Hypertension
Because of the potency of noradrenaline and the varying response to pressor substances, the possibility always exists that dangerously high blood pressure may be produced with overdoses of this pressor agent. It is desirable, therefore, to record the blood pressure every two minutes from the time administration is started until the desired blood pressure is obtained, then every five minutes if administration is to be continued. The rate of flow must be watched constantly, and the patient should never be left unattended while receiving noradrenaline. Headache may be a symptom of hypertension due to overdosage.

Site of Infusion
Whenever possible, infusions of noradrenaline should be given into a large vein, particularly an antecubital vein because, when administered into this vein, the risk of necrosis of the overlying skin from prolonged vasoconstriction is apparently very slight. Some authors have indicated that the femoral vein is also an acceptable route of administration. A catheter tie-in technique should be avoided, if possible, since the obstruction to blood flow around the tubing may cause stasis and increased local concentration of noradrenaline. Occlusive vascular diseases (for example, atherosclerosis, arteriosclerosis, diabetic endarteritis, Buerger's disease) are more likely to occur in the lower than in the upper extremity. Therefore, one should avoid the veins of the leg in elderly patients or in those suffering from such disorders. Gangrene has been reported in a lower extremity when infusions of noradrenaline were given in an ankle vein.

Extravasation
The infusion site should be checked frequently for free flow. Care should be taken to avoid extravasation of noradrenaline into the tissues, as local necrosis might ensue due to the vasoconstrictive action of the drug. Blanching along the course of the infused vein, sometimes without obvious extravasation, has been attributed to vasa vasorum constriction with increased permeability of the vein wall, permitting some leakage. This also may progress on rare occasions to superficial slough, particularly during infusion into leg veins in elderly patients or in those suffering from obliterative vascular disease. Hence, if blanching occurs, consideration should be given to the advisability of changing the infusion site at intervals to allow the effects of local vasoconstriction to subside.

Antidote for Extravasation Ischemia
To prevent sloughing and necrosis in areas in which extravasation has taken place, the area should be infiltrated as soon as possible with 10 mL to 15 mL of saline solution containing from 5 mg to 10 mg of phentolamine, an

adrenergic blocking agent. A syringe with a fine hypodermic needle should be used, with the solution being infiltrated liberally throughout the area, which is easily identified by its cold, hard and pallid appearance. Sympathetic blockage with phentolamine causes immediate and conspicuous local hyperaemic changes if the area is infiltrated within 12 hours. Therefore, phentolamine should be given as soon as possible after the extravasation is noted.

INTERACTION WITH OTHER MEDICAMENTS
Cyclopropane and halothane anaesthetics increase cardiac autonomic irritability and therefore seem to sensitise the myocardium to the action of intravenously administered adrenaline or noradrenaline. Hence, the use of noradrenaline during cyclopropane and halothane anaesthesia is generally considered contraindicated because of the risk of producing ventricular tachycardia or fibrillation. The same type of cardiac arrhythmias may result from the use of noradrenaline in patients with profound hypoxia or hypercarbia. Noradrenaline should be used with extreme caution in patients receiving monoamine oxidase inhibitors (MAOI) or antidepressants of the triptyline or imipramine types, because severe, prolonged hypertension may result.

Incompatibilities
Noradrenaline must not be mixed with other medicinal products except those mentioned in Recommended Dosage. Infusion solutions containing noradrenaline tartrate have been reported to be incompatible with the following substances: alkalis and oxidizing agents, barbiturates, chlorpheniramine, chlorothiazide, nitrofurantoin, novobiocin, phenytoin, sodium bicarbonate, sodium iodide, streptomycin.

PREGNANCY AND LACTATION
Pregnancy
Animal reproduction studies have not been conducted with noradrenaline. It is also not known whether noradrenaline can cause foetal harm when administered to a pregnant woman or can affect reproduction capacity. Noradrenaline should be given to a pregnant woman only if clearly needed.

Lactation
It is not known whether this drug is excreted in human milk. Because many drugs are excreted in human milk, caution should be exercised when noradrenaline is administered to a nursing woman.

ADVERSE EFFECTS
The following reactions can occur:

Body as A Whole: Ischaemic injury due to potent vasoconstrictor action and tissue hypoxia.
Cardiovascular System: Bradycardia (probably as a reflex result of a rise in blood pressure), arrhythmias and cardiogenic shock.
Cardiac disorders: Stress cardiomyopathy (frequency ‘not known’).
Nervous System: Anxiety, transient headache.
Respiratory System: Respiratory difficulty.
Skin and Appendages: Extravasation necrosis at injection site.

Prolonged administration of any potent vasopressor may result in plasma volume depletion which should be continuously corrected by appropriate fluid and electrolyte replacement therapy. If plasma volumes are not corrected, hypotension may recur when noradrenaline is discontinued, or blood pressure may be maintained at the risk of severe peripheral and visceral vasoconstriction (e.g. decreased renal perfusion) with diminution in blood flow and tissue perfusion with subsequent tissue hypoxia and lactic acidosis and possible ischaemic injury Gangrene of extremities has been rarely reported. Overdoses or conventional doses in hypersensitive person (e.g. hyperthyroid patients) can cause severe hypertension with violent headache, photophobia, stabbing retrosternal pain, pallor, intense sweating, and vomiting.

SYMPTOMS AND TREATMENT FOR OVERDOSE
Overdosage with noradrenaline may result in severe hypertension, reflex bradycardia, marked increase in peripheral resistance, and decreased cardiac output. These may be accompanied by violent headache, pulmonary edema, photophobia, retrosternal pain, pallor, intense sweating and vomiting. Stress cardiomyopathy was also reported in the context norepinephrine overdose. In case of accidental overdosage, as evidenced by excessive blood pressure elevation, discontinue noradrenaline **and appropriate corrective** measures initiated until the condition of the patient sabilizes.

STORAGE CONDITIONS
Store below 30°C.
Do not freeze. Protect from light.

Retain in carton till time of use. The solution should be clear, colourless when diluted.

If not required immediately, the diluted solution may be stored up to 24 hours at temperature below 30°C after mixing with the compatible solutions.

SHELF LIFE
24 months.

DOSAGE FORMS AND PACKAGING AVAILABLE
10 x 4mL ampoule (amber).
10 ampoules fixed on a PVC tray and packed into carton box.

PRODUCT	REGISTRATION	HOLDER
/MANUFACTURER		
Pharmaniaga Lifescience Sdn Bhd (198201002939)		
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47100 Puchong, Selangor Darul Ehsan, Malaysia		

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ARTWORK LOG		
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01	23.08.19	- Replace PM code to 2003255.
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